

## **Point Operators: Pixel transforms, Color transforms, Histogram equalization**

### **1. Point Operators (Pixel-wise Operations)**

Point operators are image processing operations in which the **output value of a pixel depends only on the corresponding input pixel value**, and **not on its neighbors**. Hence, these operators are **also called pixel-wise transformations**.

Mathematically, a point operator is defined as:

$$g(x, y) = T(f(x, y))$$

where:

- $f(x, y)$  → input image pixel value
- $g(x, y)$  → output image pixel value
- $T(\cdot)$  → transformation function applied independently to each pixel

Point operators are mainly used for:

- Image enhancement
- Contrast adjustment
- Brightness correction
- Color manipulation

They are **computationally efficient** and widely used in real-time systems.

Point operations are often used to change the grayscale range and distribution. The concept of point operation is to map every pixel onto a new image with a predefined transformation function.

### **2. Pixel Transforms**

#### **Definition**

A general image processing operator is a function that takes one or more input images and produces an output image.

$$g(x) = h(f(x))$$

**Pixel transforms** modify the **intensity value of individual pixels** based on a predefined mathematical function, without considering spatial relationships.

General form:

where:  $\rightarrow$  input pixel intensity ,  $\rightarrow$  output pixel intensity

## **2.1 Linear Pixel Transforms**

### **(a) Identity Transformation**

- No change in the image
- Used for reference or pipeline testing

### **(b) Image Negative**

where  $L$  is the number of gray levels (e.g., 256).

#### **Purpose:**

- Enhances white or gray details in dark regions
- Used in medical imaging (X-rays, MRI)

### **(c) Brightness Adjustment**

- $b > 0$  : increases brightness
- $b < 0$  : decreases brightness

Used in preprocessing under varying illumination conditions.

### **(d) Contrast Stretching**

- Expands or compresses the range of pixel values
- Improves visibility of details

## **2.2 Non-Linear Pixel Transforms**

### **(a) Logarithmic Transformation**

**Advantages:**

- Expands low-intensity values
- Compresses high-intensity values

**Applications:**

- Fourier spectrum visualization
- Medical and astronomical images

**(b) Power-Law (Gamma) Transformation**

- : enhances darker regions
- : enhances brighter regions

Used in **gamma correction** for display devices.

**(c) Thresholding**

Converts grayscale images into **binary images** for segmentation.

**3. Color Transforms****Definition**

**Color transforms** modify the **color components** of pixels, usually by changing the **color space** or by applying transformations to each channel.

### 3.1 RGB Color Transformations

In RGB images, each pixel is represented as:

#### Examples:

- Channel-wise enhancement
- Color inversion
- Color balancing (white balance)

#### Limitations:

- RGB is highly correlated
- Not perceptually uniform

### 3.2 Color Space Transformations

#### (a) RGB $\rightarrow$ Grayscale

Used when color information is unnecessary.

#### (b) RGB $\rightarrow$ HSV / HSI

- **Hue (H):** color type
- **Saturation (S):** color purity
- **Value/Intensity (V/I):** brightness

#### Advantages:

- Separates intensity from color
- Useful in object tracking and segmentation

#### (c) RGB $\rightarrow$ YCbCr

- : luminance
- : chrominance

Widely used in **image and video compression** (JPEG, MPEG).

### 3.3 False Color Transformations

- Assign colors to grayscale intensities
- Enhances visual interpretation

Used in:

- Remote sensing
- Medical imaging
- Thermal images

## 4. Histogram Equalization

### Definition

**Histogram Equalization (HE)** is a point processing technique used to **enhance image contrast** by **redistributing intensity values** so that the histogram becomes approximately uniform.

### 4.1 Image Histogram

The histogram represents:

- : number of pixels with intensity
- Probability of occurrence:

where is the total number of pixels.

### 4.2 Principle of Histogram Equalization

The transformation function is defined using the **cumulative distribution function (CDF)**:

where:

- $\rightarrow$  number of gray levels
- $\rightarrow$  new intensity value

This spreads intensity values over the full dynamic range.

### 4.3 Steps in Histogram Equalization

1. Compute the histogram of the image
2. Normalize the histogram to obtain probabilities

3. Compute the cumulative distribution function (CDF)
4. Map old intensities to new values using the CDF
5. Generate the enhanced image

#### **4.4 Advantages**

- Improves global contrast
- Enhances hidden details
- Simple and fast

#### **4.5 Limitations**

- May over-enhance noise
- Not suitable for images with non-uniform lighting
- Can distort natural appearance

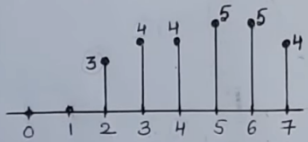
#### **Example 1:**

## HISTOGRAM EQUALIZATION → Image Enhancement.

- Graphical Representation → Data.
- Image Processing → Data related to the Digital Image.
- Representation → Frequency of Occurrence of Various gray levels.

|   |   |   |   |   |
|---|---|---|---|---|
| 6 | 6 | 7 | 7 | 6 |
| 5 | 2 | 2 | 3 | 4 |
| 3 | 3 | 4 | 4 | 5 |
| 5 | 7 | 3 | 6 | 2 |
| 7 | 6 | 5 | 5 | 4 |

0 → 0  
1 → 0  
2 → 3  
3 → 4  
4 → 4  
5 → 5  
6 → 5  
7 → 4



- Used → Manipulating Contrast & Brightness.
- Quality → Normalizing → Histogram → Flat profile

Example:

